Math 8/9 - Powers and Exponents Practice Test

Rubric									
Outcome	IE	Emerging		Developing		Proficient		Extending	
Understanding and Solving									
Mathematical Communication									

St. George's School - Academic Integrity

Academic Dishonesty includes any conduct with the intent to gain an unfair advantage in connection with an academic assessment. Academic Dishonesty can occur in many ways. Common forms of Academic Dishonesty include, but are not limited to, cheating, falsification, plagiarism, and tampering. Cheating occurs when an individual undermines the integrity of an assessment (including homework and other assignments, reports, projects, quizzes, tests, exams, or other forms of performance evaluations).

Examples of cheating include:

- Copying any part of an assessment;
- Allowing others to copy any part of an assessment;
- Improperly giving or receiving assessment information;
- Using unauthorized resources for or during an assessment;
- Skipping classes to avoid an assessment

Name : _____

Teacher's name: _____

Emerging Level

1. Write the following as a power.

   (a) $(-2) \times (-2) \times (-2)$

   (b) $\left(\frac{4}{5}\right) \times \left(\frac{4}{5}\right) \times \left(\frac{4}{5}\right) \times \left(\frac{4}{5}\right)$

2. Evaluate the following.

   (a) 3^3

   (b) $2^3 \times 7$

   (c) $\sqrt{\frac{4}{25}}$

   (d) $\sqrt[3]{-\frac{8}{27}}$

3. Express the following numbers as a product of primes.

   (a) 550

   (b) 72

4. Simplify the following using exponent laws. Do not evaluate.

   (a) $(x^6)^9$

   (b) $y^6 \div y^3$

   (c) $z^9 \times z^7$

   (d) $(-x)^0$

Developing Level

5. Evaluate the following.

   (a) $6^2 \times 2$

   (b) $3 + 5^2 \times 2$

   (c) $\frac{1}{2}(4+2)^2 - 6$

   (d) $(-3-7)^2$

6. Estimate the following to one decimal place.

   (a) $\sqrt{95}$

   (b) $\sqrt{60}$

7. Insert $<, >,$ or $=$ to create a true statement.

   (a) $-1^7 \qquad (-1)^{20}$

   (b) $(-2)^5 \qquad -2^5$

   (c) $\left(-\frac{1}{2}\right)^2 \qquad \left(-\frac{1}{2}\right)^4$

   (d) $\left(-\frac{3}{2}\right)^3 \qquad \left(-\frac{3}{2}\right)^5$

8. Simplify the following.

   (a) $\sqrt{9x^2y^6}$

   (b) $\sqrt[3]{64x^{12}y^{15}}$

Proficient Level

9. Simplify the following using exponent laws.

   (a) $(4ab)^2$

   (b) $4ab \times 5a^2b^3$

   (c) $x^{6+a} \cdot x^{3+a}$

   (d) $\dfrac{10x^7}{20x^{2a}}$

10. Evaluate the following

    (a) $4 \times (3+2)^2 + (-3)^3 - 4$

    (b) $\dfrac{10 - (4-5)^3}{5^2 - 5 \times 2^2}$

11. Find $\sqrt{176400}$, given that $176400 = 2^4 \cdot 3^2 \cdot 5^2 \cdot 7^2$
    The answer can be expressed as a product of primes or as a single number.

Extending Level

12. Simplify the following using exponent laws. Leave answers with no negative exponents.

    (a) $(-3a^3b^4)^2 \times (2a^4b^2)^3$

    (b) $\dfrac{3a^3b^5 \times 4a^4b^6}{(2a^4b^3)^3}$

13. Simplify the following using exponent laws. Leave answers with no negative exponents.

    (a) $(-3a^2b)^{-3}$

    (b) $\left(\dfrac{4a^2bc}{6a^3b^4}\right)^{-2}$

14. Simplify the following using exponent laws.

(a) $27^{x+1} \times 81^{x-3}$

(b) $\dfrac{16^{2x+1}}{32^{x-1}}$

15. Solve for x.

(a) $4x^2 - 7 = 29$

(b) $2x^3 + 10 = -6$

Extending* Level

16. Answer the following parts.

    (a) **Warm-Up:** What must be true of the values of a and b if the following equation is true.

    $$23^a = 23^b$$

    (b) **Warm-Up:** What value of x makes the following equation true?

    $$10^{2x+4} = 10^{13-x}$$

    (c) Solve for x in the following.

    $$16^{x+1} = 32^{2-x}$$